R.M. HIMALSHA JAYAMAL RATHNAYAKA

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LANGUAGES

Sinhala

English

EDUCATION

HD in Electrical and Electronic Engineering
Liverpool John Moores University (via ICBT Campus, Colombo, Sri Lanka)
October 2023 - August 2025

BEng in Electrical and Electronic Engineering
University of Sunderland (via ICBT Campus, Colombo, Sri Lanka)
January 2026 — In Progress

BSc in Natural Science
Open University of Sri Lanka
April 2024 — In Progress

SKILLS

- Project Management
- Teamwork
- Time Management
- Leadership
- Critical Thinking
- Adaptability

ABOUT ME

Higher Diploma graduate in Electrical and Electronic Engineering with Distinction and Academic Excellence Award recipient, currently completing a BEng in Electrical and Electronic Engineering at the University of Sunderland. Experience in SystemVerilog RTL design, FPGA-targeted simulation, and embedded system integration through academic and independent projects. Independently developing a custom 8-bit processor with an adaptive cache controller in Vivado XSim featuring multiple cache management policies and RTL-level guardrail mechanisms. Experienced in ROS 2 robotics integration, industrial electrical systems, and directed/constrained-random simulation workflows. Seeking entry-level RTL, FPGA, or digital design opportunities.

ACADEMIC PROJECTS

Semantic-Aware SLAM Robot with Reinforcement Learning and Triple-Layer Safety

- Multi-sensor fusion using LiDAR, IMU, RGB camera, and odometry
- Three navigation approaches designed and compared — traditional SLAM with path planning, end-to-end RL navigation, and hybrid SLAM-RL
- RL agents implemented using Stable-Baselines3 with PPO and SAC algorithms
- SLAM using Google Cartographer within ROS 2 framework
- Simulation and testing in Gazebo
- Triple-layer safety mechanism with real-time monitoring, fault detection, and recovery

Adaptive Cache Controller on Custom 8-bit Processor (In Progress)

- Designed a custom 8-bit processor architecture in SystemVerilog including ALU, register file, instruction decode, control logic, and memory interface
- Implemented adaptive cache controller supporting multiple replacement and insertion policies including RRIP-style, DRRIP-style, SHiP-style, bypass, prefetch, and Q-learning-based adaptation
- Developed hardware-compatible reinforcement learning controller with bounded state/action space and LFSR-based exploration
- Implemented hardware guardrail mechanisms including oscillation clamp, regression fallback, bandwidth cap, and conflict override
- Verified integrated RTL design using directed and constrained-random simulation methodologies
- Executed more than 1,200 verification test cases using Vivado XSim behavioural simulation environment
- Prepared architecture for FPGA-targeted synthesis and timing analysis

WORK EXPERIENCE

Electrical Engineering Intern
Hayleys Fabric PLC — Neboda, Sri Lanka

August 2025 – February 2026

- Configured and tested Schneider Electric Variable Frequency Drives (VFDs), including acceleration/deceleration profiles and operational frequency limits for industrial motor systems
- Worked with industrial instrumentation systems including PT100 temperature sensors, pressure sensors, relays, timers, and pneumatic actuators in live manufacturing environments
- Assisted in troubleshooting and maintenance of DOL, star-delta, and forward-reverse motor control systems
- Performed maintenance and fault rectification of pneumatic systems including solenoid valve replacement and leakage correction
- Conducted process-performance analysis for Stenter machine optimisation using Excel-based data analysis, contributing to production speed improvement from 16 m/min to 22 m/min and approximately 5°C operational temperature reduction
- Prepared MTTR and MTBF maintenance reliability reports using operational maintenance data
- Supported installation and commissioning activities for industrial washing and dryer systems including sensor calibration, cable routing, and electrical integration

ACADEMIC PROJECTS

1. **Water Level Controller**
- Designed a digital pump control system using logic gates and simulated float switches in Proteus, with manual override functionality.

2. **Analog Electronics – Rectifier & Amplifier Design**
- Built a full-wave rectifier with ripple filtering and a common-emitter amplifier; analyzed gain using Multisim and oscilloscope measurements.

3. **Autonomous Guided Vehicle (AGV) – Garment Industry**
- Proposed an Arduino-based AGV with stop logic using LDR sensors; developed prototype control logic for material transport automation.

4. **Apnea ECG Signal Analysis**
- Processed noisy ECG signals in MATLAB using FFT and digital filters to extract and analyze biomedical signal characteristics.

5. **Soil Moisture Monitoring System**
- Developed a PIC-based control system to sense soil moisture using ADC and activate a pump; displayed moisture levels on an LCD.

6. **Ball-on-Beam Control System (Simulation)**
- Modeled and simulated a dynamic ball-on-beam system in Simulink; applied PID control to evaluate stability and response.

7. **Reactive Circuit Analysis (RLC/RC/RL)**
- Conducted transient and steady-state analysis of reactive circuits using MATLAB, Multisim, and manual calculations.

8. **Statistical Dataset Analysis**
- Used SPSS to perform correlation, regression, and hypothesis testing on a 50-variable dataset in electrical engineering.

9. **Hospital Alert Priority Encoder (Simulation)**
- Designed and implemented a 10:4 priority encoder in Verilog; displayed highest-priority signal on a 7-segment display using FPGA simulation.

10. **PLC-Based Metal Shaping Machine (Simulation)**
- Simulated a metal shaping process using ladder logic and pneumatic sequencing in Automation Studio.

TECHNICAL SKILLS

Hardware Description Languages: SystemVerilog, Verilog
Simulation and EDA Tools: Vivado XSim (AMD/Xilinx), Xilinx ISE, Proteus, MATLAB Simulink
Robotics and Embedded Frameworks: ROS 2, Gazebo, Stable-Baselines3, Arduino IDE
Verification Methodology: Directed testbench, constrained-random simulation, functional coverage, hard-gate acceptance testing
Programming: Python (verification scripting, data analysis), C/C++, MATLAB
Digital Design: RTL design, combinational and sequential logic, finite state machines, cache architecture, processor pipeline design
Industrial Systems: Schneider Electric VFDs, PLC ladder logic, motor control circuits, industrial sensor integration